

Design Thinking Board

AI-Based Health Monitoring Wristband

Empathy

- Adults (30+)
- Busy professionals
- Ignore early symptoms
- Feel anxious about heart health
- Expensive checkups

Problem

How might we provide early, real-time detection of heart attack risks using an affordable wearable device?

Define

- Need continuous monitoring
- Need early alerts
- Need simple wearable solution
- Reduce hospital dependency

Testing

- Test on 10–15 users
- Compare with ECG
- Check accuracy
- Improve system

Prototype

Components:

- MAX30100
- AD8232
- ESP32
- Firebase

Working:

- Collect data
- Send to app
- AI analysis
- Show risk (Safe/Warning/Danger)

Ideate

- Smart wristband
- Mobile app
- AI prediction
- Emergency alerts
- Voice alerts

Final Solution

AI-based wristband that monitors heart activity in real-time and alerts users about potential risks.

Impact:

Prevents sudden heart attacks through early detection.